

Md. Abid Chowdhury

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Dhaka, Bangladesh

EDUCATION

- **B.Sc in Robotics & Mechatronics Engineering** 2022 - 2026
University of Dhaka Dhaka, Bangladesh
CGPA: 3.3/4.00, Till 7th Semester
Contact: rme@du.ac.bd
Thesis title: Bat-inspired Passive Locking Claw and Leg Design and Development for UAVs.

WORK EXPERIENCE

- **Cortex AI Lab** Dhaka, Bangladesh
Researcher (Director: Shifat E. Arman) Feb 2024 - Present
 - Conduct advanced research across multidisciplinary fields of Robotics and Mechatronics, integrating state-of-the-art Machine Learning, Computer Vision, and bio-inspired artificial intelligence frameworks.
- **Mechatronics and Artificial Intelligence in Medicine (MAIM) Lab** Dhaka, Bangladesh
Undergraduate Research Assistant (Director: Dr. Abhishek Kumar Ghosh) Nov 2024 - Mar 2025
 - Served as a researcher for the "Flexipede: A Bio-Inspired, Modular Myriapod Robot for Rough-Terrain Traversal" project, managing simulation, fabrication, and structural validation.
- **University of Dhaka** Dhaka, Bangladesh
Undergraduate Research Assistant (Supervisors: Dr. Shamim Ahmed Deowan, Shifat E. Arman) Sep 2024 - Present
 - Served as a researcher for a turtle-inspired modular swimming robot utilizing a minimal actuator configuration for high-efficiency underwater exploration.

RESEARCH PROJECTS

- **Bat-Inspired Passive Locking Claw and Leg Design and Development for UAVs**
Supervisors: Dr. Shamim Ahmed Deowan, Shifat E. Arman Jan 2026 - Present
 - Simulated and validated a designed and prototyped bio-inspired passive locking and active unlocking perching system to enable UAVs to persistently hang onto cylindrical structures for extended data collection.
 - Formulated an analytical mathematical framework validated through structural FEA (Static Structural) loops.
- **Flexipede (Decipede): A Bio-Inspired, Modular Myriapod Robot for Rough-Terrain Traversal**
Supervisor: Dr. Abhishek Kumar Ghosh Jul 2024 - Dec 2025
 - Aided in engineering a mechanically intelligent modular robot architecture mimicking myriapod kinematics, featuring a customized single-actuator bipedal walking mechanism.
 - Aided in the fabrication of the robot and later data collection and analysis of that data.
- **Turtle-Inspired Modular Swimming Robot with a Minimal Number of Actuators**
Supervisors: Shifat E. Arman, Dr. Shamim Ahmed Deowan Sep 2024 - Present
 - Executed coupled Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) simulations to optimize multi-material hydrodynamics and pressure distribution constraints.

- Aided in developing kinematic predictions in Webots simulation environments and verified the framework through empirical performance trials on full-scale physical hardware.

Burst Pressure Enhancement of Steel Pipelines Using Variable-

• **Angle Composite Overwraps: A Numerical Study**

Supervisor: Dr. Md. Abdus Shabur

Oct 2025 - Feb 2026

- Modeled the stress and load-sharing dynamics of 4 distinct industrial composite material systems used for global reinforcement and structural rehabilitation of transmission piping networks.
- Engineered a validated numerical simulation framework inside ANSYS ACP (Pre) and Static Structural to isolate optimal anisotropic fiber winding angles under extreme hydrostatic loading conditions.

Numerical Analysis of Burst Pressure Improvement in Corroded

• **Steel Pipelines Using Variable-Angle Composite Overwraps**

Supervisor: Dr. Md. Abdus Shabur

Oct 2025 - Feb 2026

- Developed a detailed 3D finite element framework analyzing structural rehabilitation efficiency for steel pipes containing non-uniform, localized corrosion thinning defects.
- Evaluated interface stress distribution and progressive deformation patterns across varied winding orientations to mathematically define safety metrics and Maximum Allowable Operating Pressure (MAOP) gains.

PUBLICATIONS

[1] Flexipede: A Bio-Inspired, Modular Myriapod Robot for Rough-Terrain Traversal.

Authors: Samudra Jit Saha, Md. Abid Chowdhury, Sayma Islam, Shamim Ahmed Deowan, Shifat E. Arman, and Abhishek K. Ghosh; **Journal:** Robotics; **Publisher:** MDPI

[2] Material Selection and Fiber Orientation Effects in Composite Overwrapped Steel Pipelines: A Numerical Study for Burst Pressure Enhancement.

Authors: Samudra Jit Saha, Marufa Akter, Nowreen Jahan, Sayma Islam, **Abid Chowdhury**, and Md. Abdus Shabur; **Journal:** Journal of Sustainable Smart Materials and Structural Systems

SKILLS

- **Programming Languages & Frameworks:** Python, C/C++, MATLAB, L^AT_EX, ROS 2 (Jazzy).
- **AI & Machine Learning:** PyTorch, TensorFlow, Keras, Hugging Face, LangChain, Scikit-learn, OpenCV, Edge Impulse, CUDA.
- **Simulation & Modeling:** Ansys, COMSOL Multiphysics, Fusion 360, KiCAD.
- **Hardware & Prototyping:** ESP32, Raspberry Pi Pico, STM32, PLC, Oscilloscope, Function Generator, 3D Printing.

ACHIEVEMENTS

- **Silver Medal (2nd Place) – Physics and Applied Engineering** 2026
8th World Innovation Competition and Exhibition (WICE) Northern University Bangladesh
- **Winner (4th Place) – Robotics & IoT** 2026
1st Global Robotics and Innovation Consortium 2026 United International University
- **Champion (1st Place) – Robotics Project Showcase** 2025
Robotronics Fest 2025 University of Dhaka
- **Research Grant – ICT Division Innovation Fund** 2024
ICT Ministry Dhaka, Bangladesh